

ROAD ACCIDENT ANALYSIS DASHBOARD

Data Analytics Documentation & Analysis Report

Prepared from: MAINDATA.xlsx & CALENDER.xlsx | Current Year (CY) 2022 vs Previous Year (PY) 2021 | 20 September 2026

1. Purpose & Scope

This report explains how the Road Accident Analysis dashboard is built from raw accident records, verifies every number on it, and analyses where, when and under what conditions people are being hurt on the road.

In short: casualties fell 11.9% in 2022 (222,146 → 195,737), but 2022 contains almost no Scottish police-force records, which overstates the fall. Like-for-like, casualties fell 5.7%. Severity did improve, and the danger is concentrated in rural roads, 60 and 70 mph limits, unlit roads at night, and weekends.

2. Data Source & Model

Item	Role	Contents
MAINDATA.xlsx	Fact table	307,973 accidents × 24 columns, one row per accident (Accident_Index is unique), 1 Jan 2021 to 31 Dec 2022 (163,554 in 2021; 144,419 in 2022).
CALENDER.xlsx	Date dimension	730 daily rows with Date, year, month. Defines CY (2022) and PY (2021) and orders the months.
Dashboard	Report page	Five KPI cards with YoY change, six vehicle cards, monthly area chart, road-type chart, two donuts (urban/rural, light) and two slicers (Junction Control, Road Surface).

The model is a star: CALENDER[Date] (one) to MAINDATA[Accident Date] (many). Key fields:

Accident_Severity (Fatal / Serious / Slight, recorded per accident), **Number_of_Casualties** (summed for every casualty KPI), **Vehicle_Type** and its group, **Road_Type**, **Speed_limit**, **Urban_or_Rural_Area**, **Light_Conditions** and its group, **Road_Surface_Conditions**, **Weather_Conditions**, **Junction_Control / Junction_Detail**, **Time** (stored as “31-12-1899 hh:mm:ss”, so the hour must be parsed), **Police_Force**, **Local_Authority** and Latitude / Longitude.

3. How the Dashboard Was Built

Reconstructed from the dashboard (Power BI terms; the same logic works in Excel or Tableau). It reproduces every figure exactly.

- **Clean:** load both files, set data types, and create the grouped columns (Vehicle_Type, Light_Conditions and Junction_Control groups). The grouping also fixes label variants such as “Auto traffic sigl”.
- **Model:** one-to-many relationship CALENDER[Date] → MAINDATA[Accident Date]; use CALENDER[month] on the month axis.
- **Measure:** CY is the latest year in the calendar and PY the year before, with no hard-coded years (below).
- **Visualize and interact:** KPI cards with YoY %, vehicle cards, area, column and donut charts, and two slicers that cross-filter every visual.

```

CY Casualties = VAR cy = CALCULATE ( MAX ( CALENDER[year] ), ALL ( CALENDER ) )
RETURN CALCULATE ( SUM ( MAINDATA[Number_of_Casualties] ), CALENDER[year] = cy )
PY Casualties = (same, with CALENDER[year] = cy - 1)
CY Fatal Casualties = CALCULATE ( [CY Casualties], MAINDATA[Accident_Severity] = "Fatal" )
CY Total Accidents = DISTINCTCOUNT ( MAINDATA[Accident_Index] ) filtered to CY
YoY % = DIVIDE ( [CY Casualties] - [PY Casualties], [PY Casualties] )

```

4. Key Metrics & Reconciliation

Metric	CY 2022	PY 2021	YoY (dashboard)	YoY (like-for-like)
Total Casualties	195,737	222,146	-11.9%	-5.7%
Total Accidents	144,419	163,554	-11.7%	-5.2%
Fatal Casualties*	2,855	4,280	-33.3%	-27.5%
Serious Casualties	27,045	32,267	-16.2%	-8.1%
Slight Casualties	165,837	185,599	-10.6%	-4.8%

* “Fatal Casualties” counts every casualty in a fatal-severity accident, not only deaths: 1,549 fatal accidents produced 2,855 casualties. Like-for-like excludes eight Scottish police-force areas (Section 5.1).

Derived metrics used below: fatal-accident rate = fatal accidents ÷ accidents (1.07% in 2022, 1.47% in 2021); KSI rate = (fatal + serious) ÷ accidents (14.1% vs 14.9%); casualties per accident (1.36 both years); risk multiple = segment fatal rate ÷ overall fatal rate.

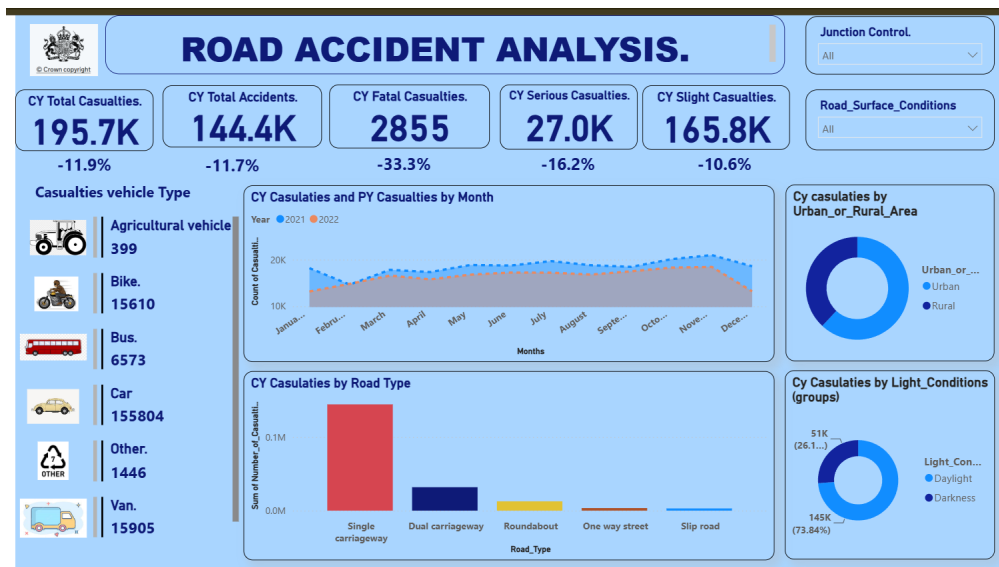


Figure 1 — The dashboard. Every card, donut and chart was recomputed from MAINDATA and matches (e.g. Car 155,804; Bike 15,610; darkness 51,198 = 26.16%).

5. Deep Analysis (CY 2022)

5.1 A coverage break behind the headline

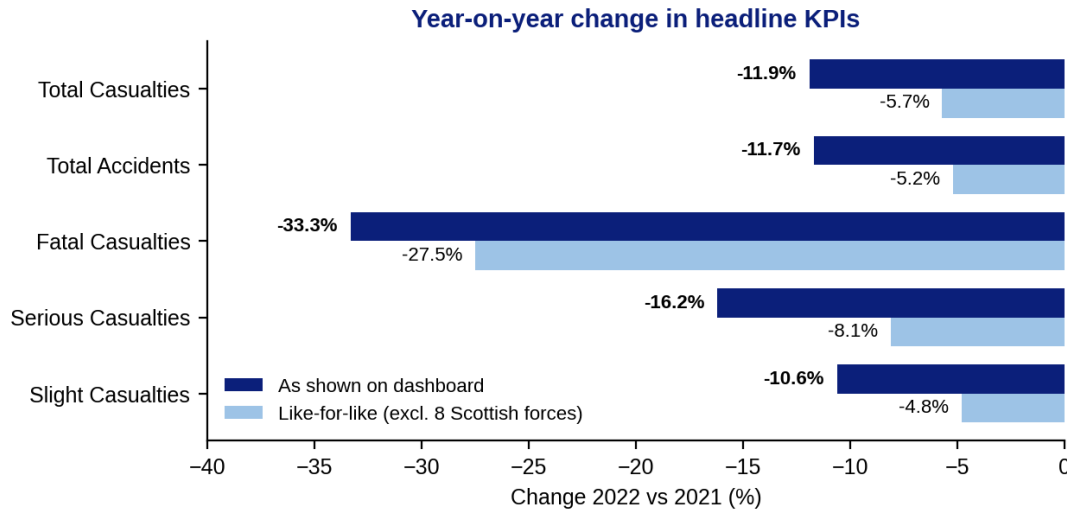


Figure 2 — Every headline KPI shows a smaller improvement once Scottish forces are excluded from both years.

Eight Scottish police-force areas have 11,537 accidents in 2021 but only 292 in 2022, and they account for **14,599 of the 26,409 fewer casualties (55%)**. Like-for-like, casualties fell 5.7%, accidents 5.2% and casualties in fatal accidents 27.5%. The fatal-accident rate still improved (1.45% to 1.07%), so **the safety gain is real, but about half the headline drop is a data-coverage effect**.

5.2 Vehicle type

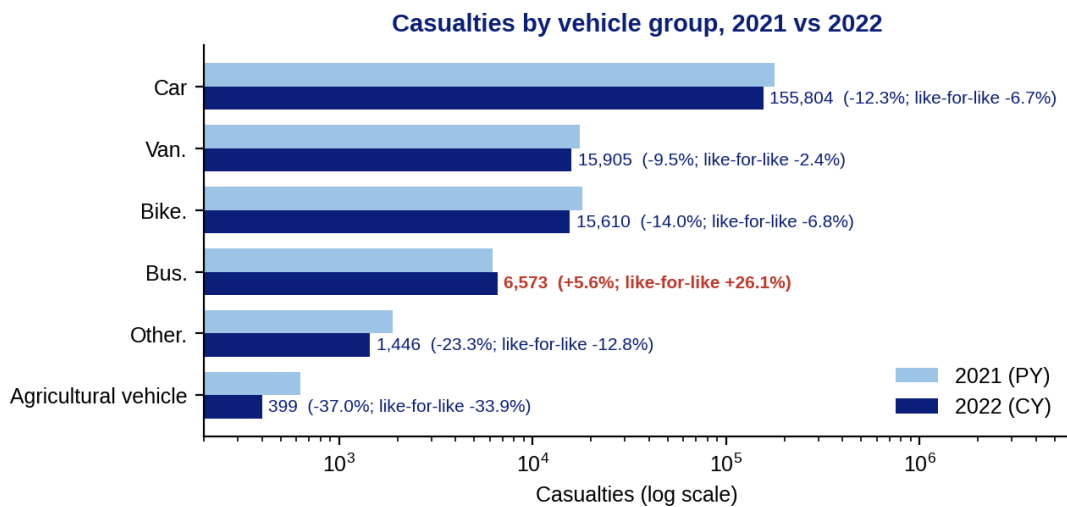


Figure 3 — Casualties by vehicle group. Bus is the only group that rose.

- Cars are 79.6% of casualties and drove 82.8% of the fall. **Bus and coach casualties rose 5.6% as shown and 26.1% like-for-like** (5,210 → 6,570), the clearest adverse trend.
- Caution: each accident carries one Vehicle_Type, so casualties are credited to that single vehicle, and “Bike” is almost all motorcycles (only 26 pedal-cycle records).

5.3 Monthly pattern

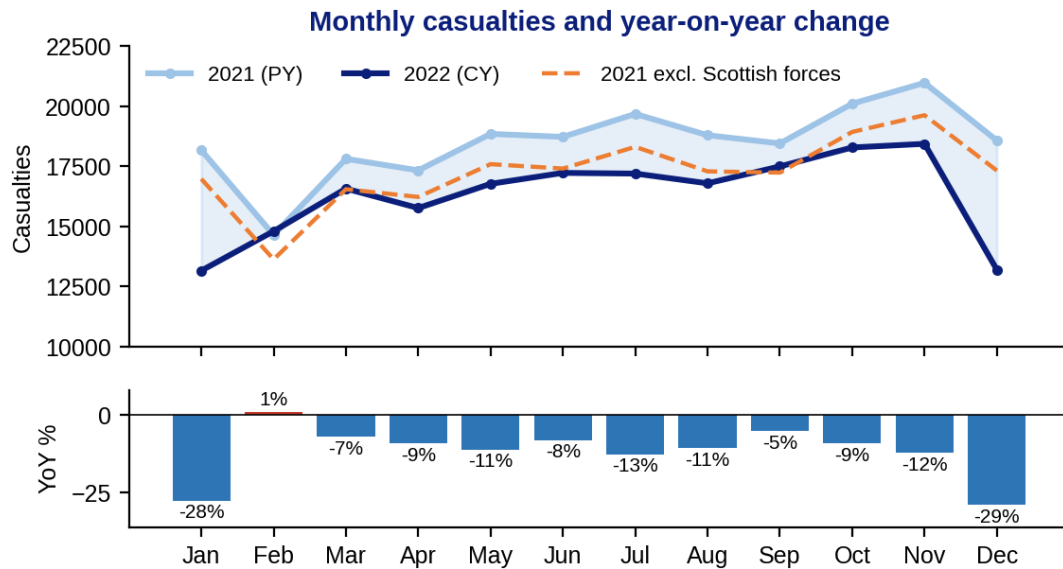


Figure 4 — Monthly casualties (dashed: 2021 excluding Scottish forces) and YoY change.

- Volume peaks in October–November. **January (-28%) and December (-29%) 2022** stand out and are not explained by Scotland (still -23% and -24% like-for-like); December 2022 averages about 310 accidents a day against 454 in November. Confirm with the data owner.

5.4 Urban vs rural

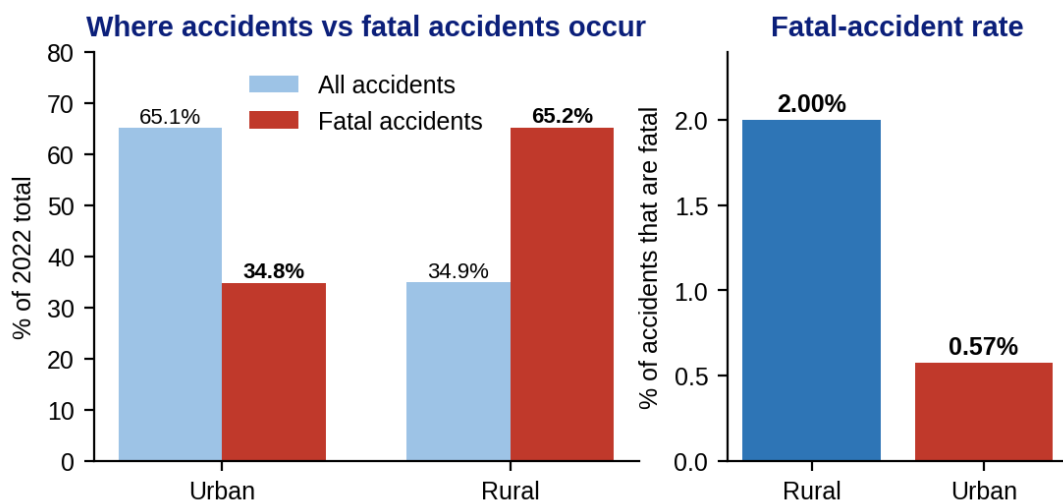


Figure 5 — Rural roads are a third of accidents but almost two-thirds of fatal accidents.

- Urban carries 61.9% of casualties, rural 38.1%. But rural roads hold 34.9% of accidents and **65.2% of fatal accidents**: a 2.00% fatal rate against 0.57% urban (3.5×). Urban is a volume problem; rural is a severity problem.

5.5 Speed limit and road type

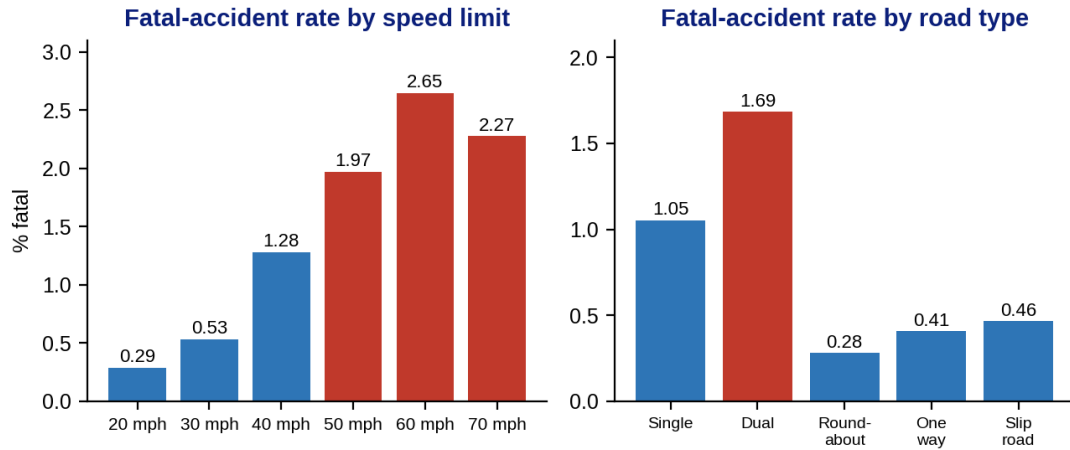


Figure 6 — Fatal-accident rate rises sharply on high-limit roads and dual carriageways.

- **60 mph roads are 14.4% of accidents but 35.4% of fatal accidents** (2.65% fatal rate, 5× the 30 mph rate). Rural roads with a 60 or 70 mph limit are 20.3% of accidents but **49.6% of fatal accidents**.
- Dual carriageways have the highest fatal rate (1.69%); roundabouts (0.28%), slip roads and one-way streets are safest. Single carriageways dominate volume (73.9% of casualties) but no traffic-volume data exists to compute risk per mile.

5.6 Light, time of day and week

- **Unlit darkness:** darkness is 25.4% of accidents but 37.8% of fatal ones. The risk sits in darkness with no street lighting: 3.83% fatal rate (4.3× daylight), 5.1% of accidents but 18.1% of fatal accidents. Lit night driving (1.06%) is barely riskier than daylight. Rural + dark + 60/70 mph has a 3.49% fatal rate.

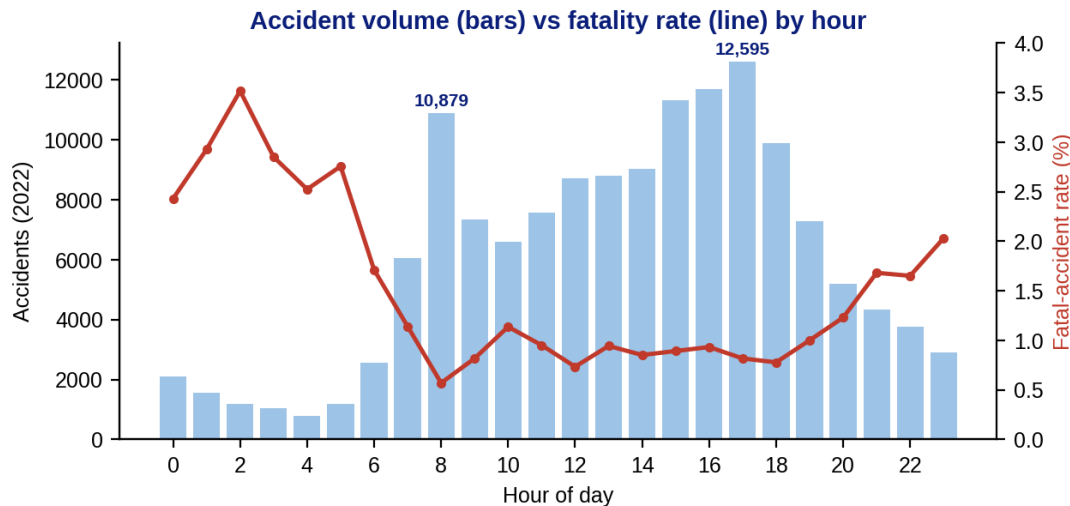


Figure 7 — Volume peaks at 08:00 and 17:00; fatality risk peaks in the early hours.

- Fatal rate is lowest at 08:00 (0.57%) and highest at 02:00 (3.52%). The window 22:00–05:59 has 10.1% of accidents but **22.1% of fatal accidents**.
- Friday is busiest (23,863 accidents), Sunday quietest but most severe (1.76% fatal rate). Saturday and Sunday are 24.3% of accidents but 34.7% of fatal ones.

5.7 Road surface and weather

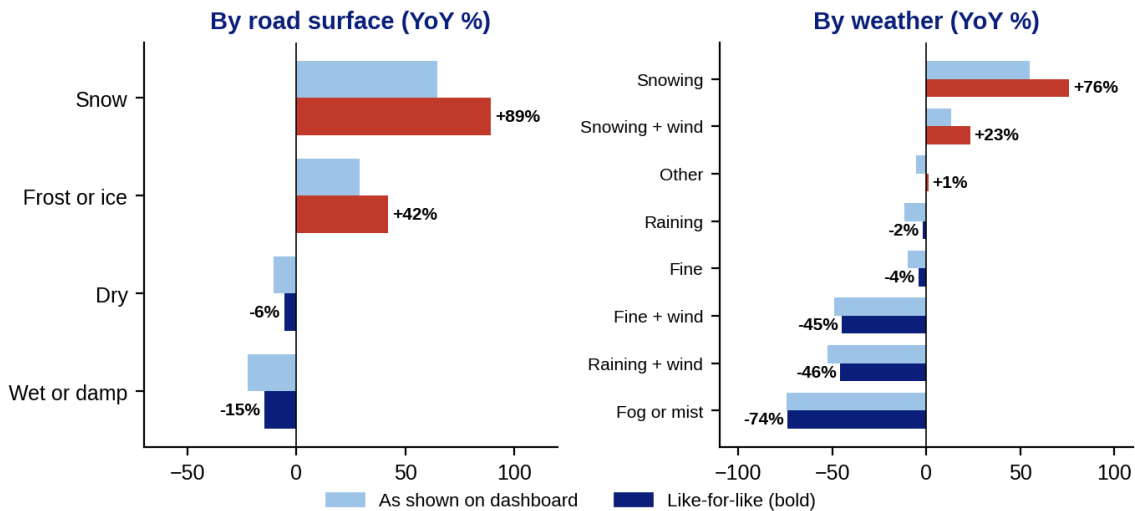


Figure 8 — Wet and foggy conditions fell; wintry conditions rose (dark bars are like-for-like).

- Most casualties happen in benign conditions (67.4% dry, 80.0% fine weather). Yet **frost/ice casualties rose 29% (42% like-for-like) and snow 65% (89%)**, while wet-surface casualties fell 22%. Adverse conditions did not raise severity (fatal rate 0.68% on snow vs 1.08% dry), consistent with drivers slowing down.

5.8 Junctions and geography

- Two in five casualties (40.0%) happen away from junctions, and those accidents are more severe (1.77% fatal vs 0.68% at T-junctions and 0.24% at roundabouts). Give way or uncontrolled junctions hold 48.3% of casualties; signalised junctions have the lowest fatal rate (0.47%).
- The Metropolitan Police area records 14.6% of casualties (28,557) but only a 0.52% fatal rate. Authorities with the highest fatal rates (at least 300 accidents) are rural or semi-rural, such as South Oxfordshire (3.80%), North Kesteven (3.71%) and Cherwell (3.37%), though counts are small (10 to 15 fatal accidents).

6. Key Findings

Observation	Detail
Headline vs reality	Casualties -11.9% as shown; -5.7% like-for-like. 55% of the drop is missing Scottish records.
Severity improved	Fatal-accident rate 1.47% → 1.07%; KSI rate 14.9% → 14.1%.
Rural severity	35% of accidents, 65% of fatal accidents; fatal rate 3.5× urban.
High-speed roads	60 mph: 14% of accidents, 35% of fatal. Rural 60/70 mph: 20% of accidents, 50% of fatal.
Unlit night roads	5.1% of accidents, 18.1% of fatal accidents (3.83% fatal rate).
Night and weekends	22:00–05:59: 10% of accidents, 22% of fatal. Sat + Sun: 24% of accidents, 35% of fatal.
Wintry conditions	Frost/ice casualties +42%, snow +89% (like-for-like), while wet and foggy conditions fell.
Bus and coach	Only vehicle group that rose: +26.1% like-for-like.

Urban and commuter-hour roads generate the volume; rural, high-speed, dark and weekend roads generate the deaths. Targeted action there will do more than a blanket campaign.

7. Data Quality & Dashboard Review

Issue	Detail	Suggested fix
Scottish coverage break	11,537 accidents in 2021 vs 292 in 2022 from eight Scottish forces.	Obtain the missing 2022 data, or add a Police_Force slicer and a coverage note.
“Fatal Casualties” label	Counts all casualties in fatal accidents (2,855), not deaths.	Rename it, and add a fatal-accident-rate KPI (1.07%).
One vehicle per accident	Casualties credited to a single Vehicle_Type.	Treat vehicle comparisons as indicative; add a vehicle-level table.
“Others.” junction bucket	32% of casualties; 99.6% are “Not at junction” per Junction_Detail.	Merge with “Not at junction or within 20 metres”.
Minor data issues	Time has a 1899 placeholder date and 17 blanks; Carriageway_Hazards is 98.2% blank; five rows have 10 or 15 mph limits.	Parse the hour, fill blanks with “None”, flag outliers.
Chart design	Raw field names in titles; monthly axis starts near 10K; legend says 2021/2022 while the title says CY/PY; unlabelled donut; five bar colours.	Use plain titles, a zero-based axis, consistent CY/PY labels, data labels and one bar colour.
Missing views	No time of day, day of week, speed limit, severity split or map.	Add an hour × weekday heatmap, speed-limit fatal rate, severity split and a Latitude/Longitude map.
No exposure data, two years only	No traffic volume; one YoY comparison.	Rates are per accident, not per mile; extend the history.

8. Recommendations

- **Target rural 60 and 70 mph roads first:** they hold about half of fatal accidents in one-fifth of accidents. Speed management, average-speed cameras, junction and bend treatments.
- **Manage unlit rural roads at night:** lighting, reflective studs and signage, or lower limits where lighting is not feasible.
- **Focus night and weekend messaging** on 22:00–05:59 and Saturday–Sunday. The data has no contributory-factor field, so check causes against other sources first.
- **Prepare for winter** (gritting, alerts) and **review bus and coach safety**, the only group with a rising trend.
- **Fix the analytics:** resolve the 2022 Scotland gap before publishing YoY claims; add casualty-level fields, contributory factors and traffic exposure; extend beyond two years; keep CY/PY measures dynamic so the dashboard rolls forward on refresh.